

Pharmacology of Opioids

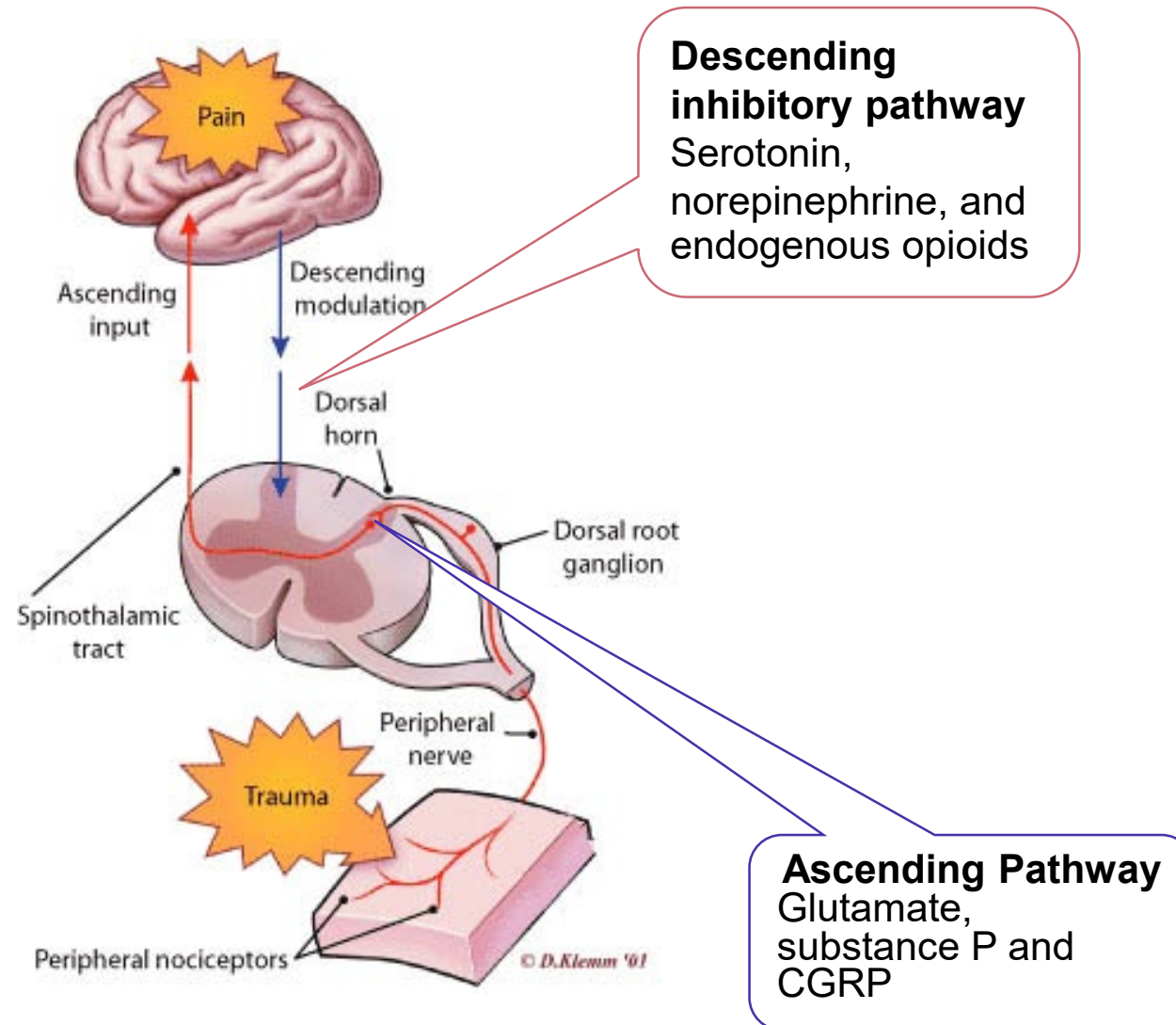


Pongsatorn Meesawatsom BPharm, PhD
Department of Pharmacology
Faculty of Pharmacy
Mahidol University

Objectives

- ★ อธิบายผลของการกระตุ้น **opioid receptor** ทั้งในเซลล์และใน **pain pathway** ได้
- ★ ยกตัวอย่างยาที่ออกฤทธิ์ต่อ **opioid receptor** แบบต่างๆได้
- ★ อธิบายผลข้างเคียงจากการใช้ **opioids** ได้
 - ◇ Opioid receptor mediated
 - ◇ Non opioid receptor mediated
- ★ อธิบายคุณสมบัติทางเภสัชวิทยาของ **opioids** ใช้บ่อยทางคลินิกได้
- ★ อธิบายอันตรกิริยาที่สำคัญของ **opioids** ได้
 - ◇ PK, PD

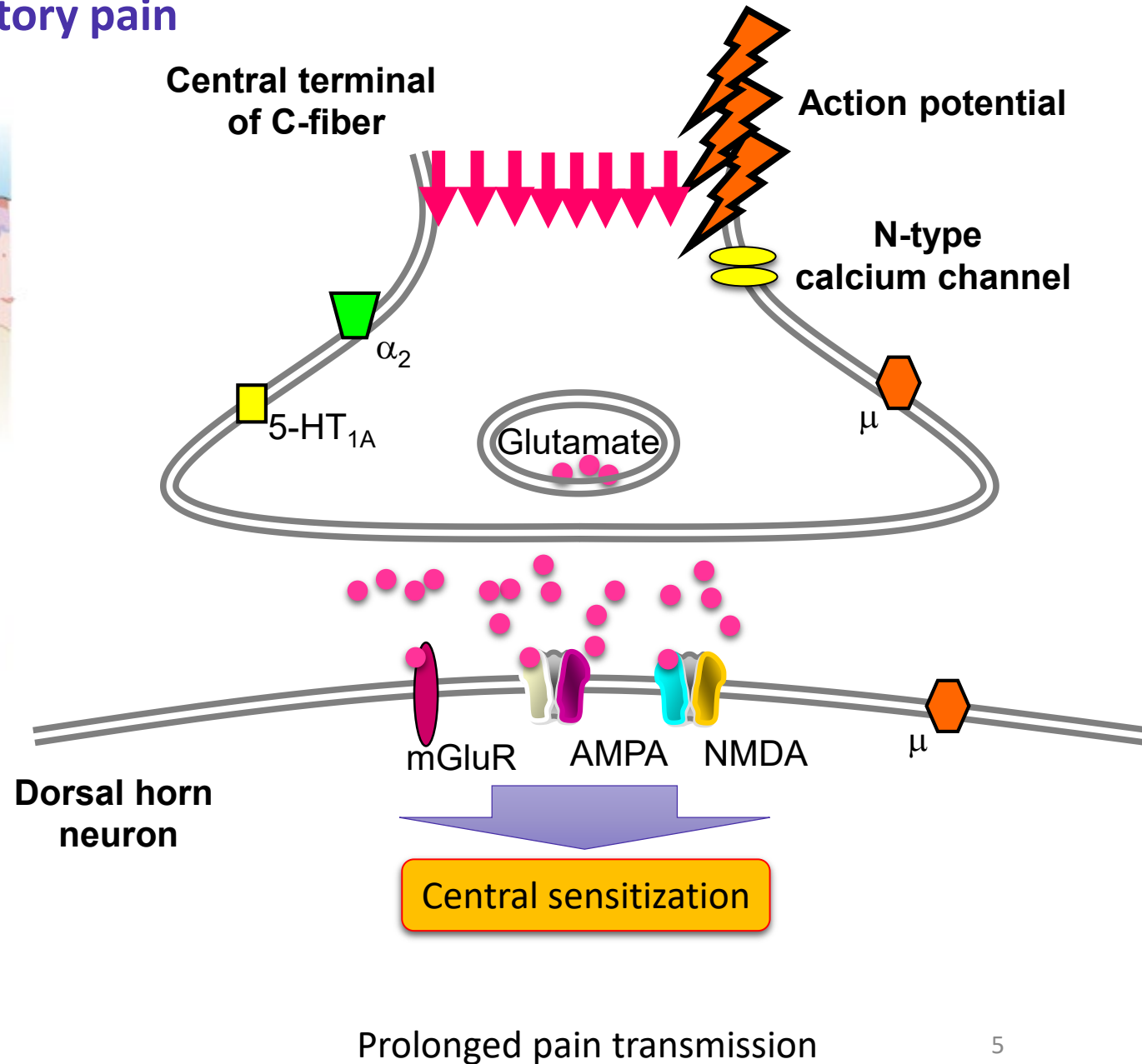
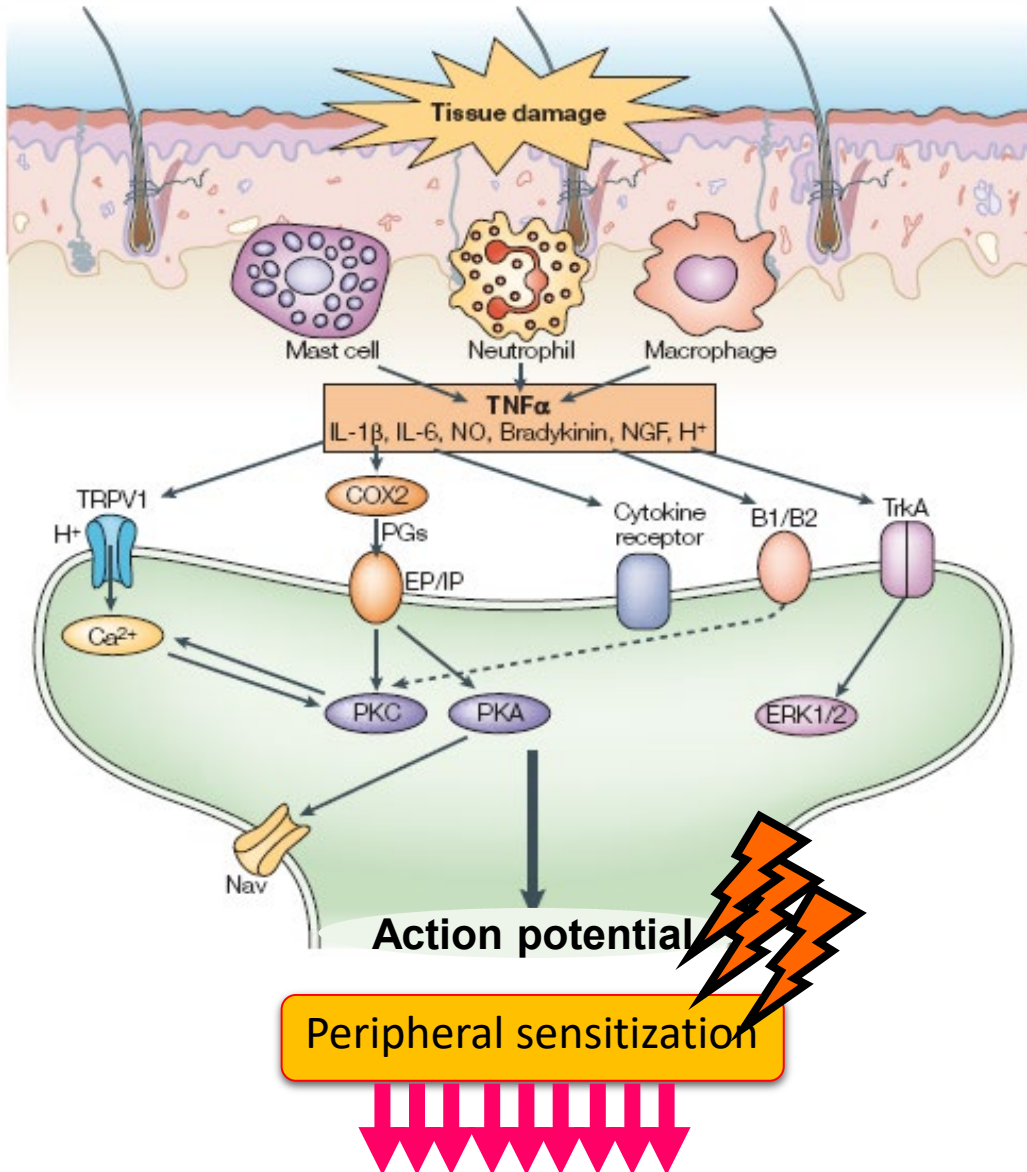
Pain pathway



Major types of pain based on mechanisms

- ★ Nociceptive/inflammatory pain
 - ◇ Activation/sensitization of nociceptor
- ★ Neuropathic pain
 - ◇ Lesion/injury to pain pathway leading to maladaptation
- ★ Nociplastic pain (dysfunctional pain)
 - ◇ Abnormal pain processing

Pain pathway sensitization in inflammatory pain



Analgesic classification

- ★ Opioid analgesics

- ★ Non-opioid analgesics

 - ◇ Paracetamol

 - ◇ NSAIDs/Coxibs

- ★ Adjuvants (coanalgesic)

 - ◇ Antidepressants

 - ◇ Antiepileptics

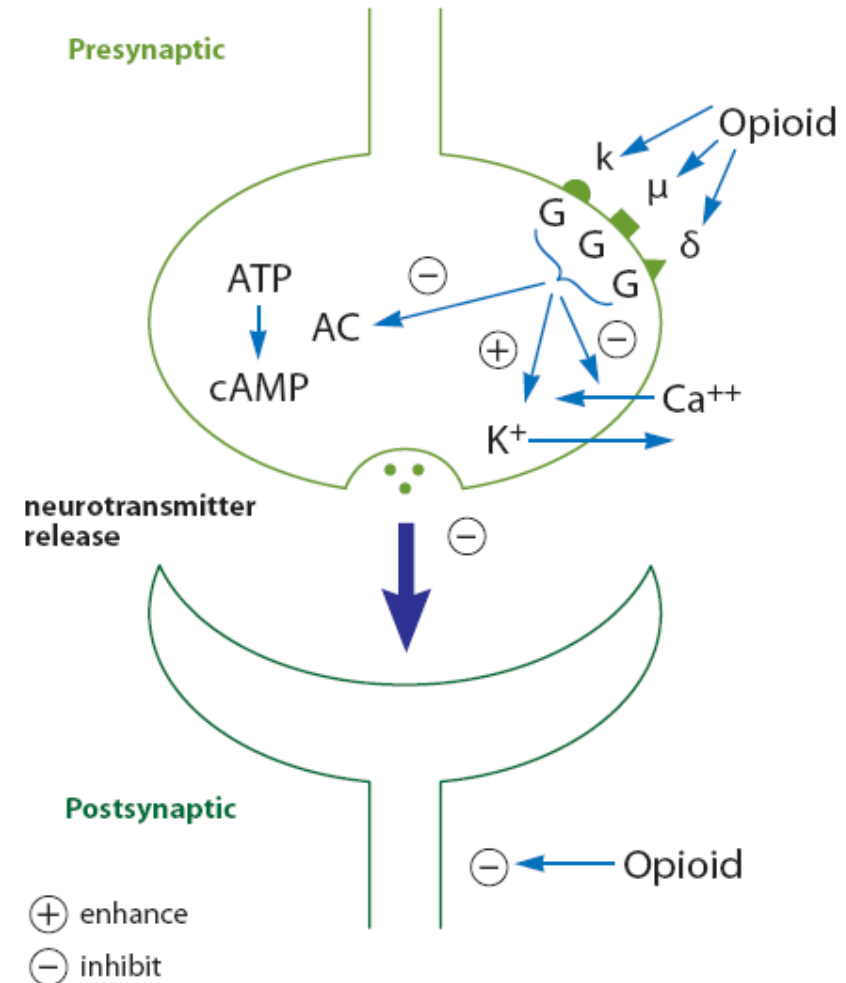
 - ◇ Muscle relaxants

 - ◇ Steroids

 - ◇ Bisphosphonates

Opioid receptors

- ★ Subtype
 - ◇ μ (MOP)
 - ◇ δ (DOP)
 - ◇ κ (KOP)
 - ◇ Opioid related nociceptin receptor 1 (OPRL1) or NOP
- ★ G-protein couple receptors coupled with G_i
 - ◇ Inhibit adenylate cyclase
 - ◇ Inhibit Na^+ and Ca^{2+} channels
 - ◇ Enhance K^+ channel



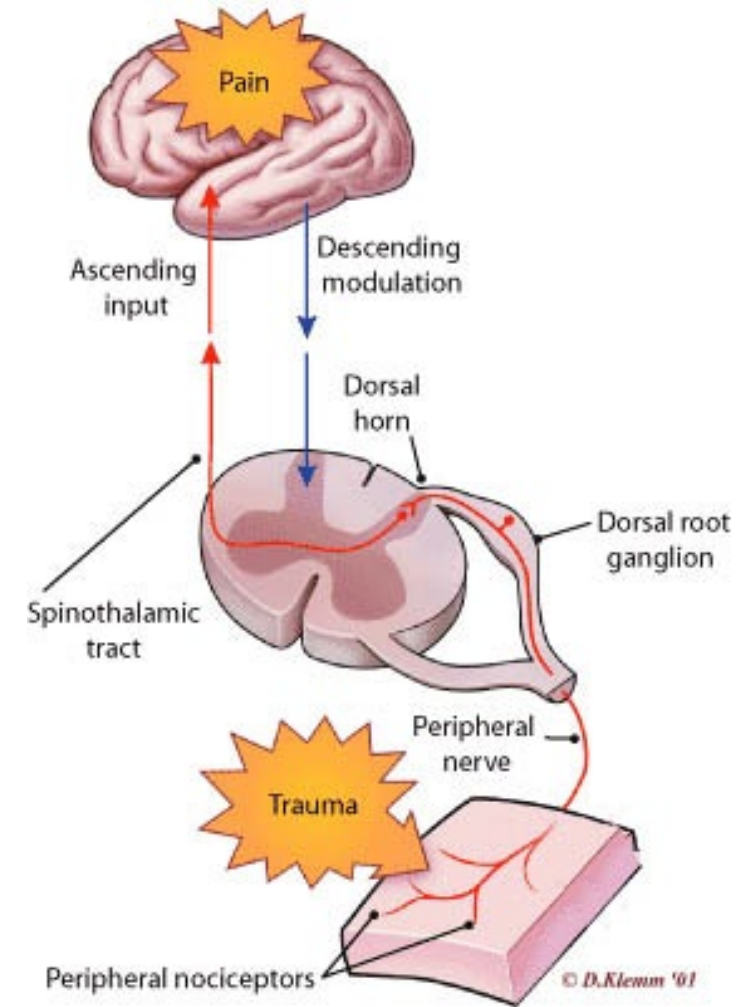
Sites of analgesic actions of opioid receptor agonists

Higher center; thalamus, somatosensory cortex

Dorsal horn cell

Central terminal of nociceptive fibers

Peripheral nociceptive terminal



Opioid receptor distribution and pharmacological effects

Receptor	μ	δ	κ	NOP
Analgesia				
▪ Supraspinal	+++	—?	—	Antiopioid
▪ Spinal	++	++	+	++
▪ Peripheral	++	—	++	—
Sedation	++	—	++	—
Respiratory depression	+++	—	—	—
Euphoria	+++	—	—	—
Dysphoria and hallucinations	—	—	+++	—
Pupil constriction	++	—	++	—
Reduced gastrointestinal motility and secretion	++	++	+	—
Physical dependence	+++	—	—	—

Drugs acting on opioid receptor

★ Weak opioids (analgesic ceiling effect)

- ◇ Codeine

- ◇ Tramadol (Tramal®)★

★ Strong opioids (no analgesic ceiling effect)

- ◇ Morphine

- ◇ Oxycodone

- ◇ Methadone★

- ◇ Meperidine (pethidine)★

- ◇ Fentanyl (Durogesic®), sufentanil, remifentanyl

- ◇ Heroin

★ Partial agonist/mixed agonist-antagonist

- ◇ Buprenorphine (Temgesic®,)

- ◇ Nalbuphine (Nulphin®)

- ◇ Pentazocin (Sosegon®)

★ Antagonist

- ◇ Naloxone (Narcan®)

- ◇ Naltrexone



With other mechanisms

Selectivity of opioid drugs for receptor subtypes

	μ	δ	κ
Weak			
▪ Tramadol, codeine	+	?	?
Strong			
▪ Morphine, oxymorphone, dextropropoxyphene	+++	+	+
▪ Methadone	+++		
▪ Meperidine (pethidine)	++	+	+
▪ Fentanyl, sufentanil	+++	+	
Partial/mixed agonist-antagonist			
▪ Nalbuphine	-		++
▪ Pentazocine	$\pm\pm$	+	++
▪ Buprenorphine	$\pm\pm\pm\pm$	-	-
Antagonists			
▪ Naloxone	---	-	---
▪ Naltrexone	---	-	---

+ agonist, $\pm$ partial agonist, - antagonist

Equianalgesic table of opioids

	Parenteral	Oral
Fentanyl	0.1	-
Oxymorphone	1	10
Hydromorphone	1.5	7.5
Oxycodone	-	20
Morphine	10	30
Hydrocodone	-	30
Meperidine	75-100	300
Tramadol	80-100	120
Codeine	100-120	200-240

Common opioid side effects

- ★ Nausea/vomiting

 - ◇ Structure related

 - ◇ Serotonin side effects

- ★ Constipation

 - ◇ **Long lasting, not tolerated**

 - ◇ Should combine laxative in chronic use of opioids

- ★ Sedation

- ★ Respiratory depression

 - ◇ Rare in appropriate use

 - ◇ Precipitating factors

 - ❖ Benzodiazepines, asthma, chronic obstructive pulmonary disease (COPD), brain trauma, CNS depressants

Less common opioid side effects

- ★ Hypotension

- ★ Opioid-induced histamine release

 - ◇ Itching

 - ◇ Hypotension

- ★ Urinary retention

- ★ Euphoria and addiction

 - ◇ Rare in appropriate use

 - ◇ Pethidine is highly addictive

- ★ Other CNS side effects

 - ◇ Miosis (not tolerated)

 - ◇ Delirium, psychosis, myoclonus jerk

Opioid tolerance and withdrawal

- ★ Long-term opioid use may cause μ receptor down regulation → tolerance
- ★ Abrupt discontinuation leads to withdrawal
- ★ The tolerance to each effect is different in the degree
- ★ The crossed tolerance is incomplete among opioids
- ★ Abruptly cessation leads to withdrawal symptoms
 - ◇ Severity $\propto$ potency, dose, duration

Drug	Route	Duration of Action (Hours)	Elimination $t_{1/2}$ (Hours)	Active Metabolites
Weak opioids				
Codeine	Oral, parenteral	4–6	3	Yes
Tramadol	Parenteral, oral	3–5	6	Yes
Strong opioids				
Morphine	Parenteral, oral	4–5	2.3	Yes
Meperidine ^b	Parenteral, oral	3–4	3	Yes
Oxycodone	Oral	3–5	3	Yes
Methadone	Oral, parenteral	4–5	23	No
Fentanyl	Parenteral	0.5–1	3.7	No
Partial Agonists and Mixed-Acting Compounds				
Buprenorphine	Parenteral, sublingual	4–6	5	No
Nalbuphine	Parenteral	4–5	5	No
Pentazocine	Parenteral, oral	3–5	4	No
Antagonists				
Naloxone	Parenteral	1–2 ^c	1	No
Naltrexone	Oral ^d	24 ^c	4	Yes

$t_{1/2}$, Half-life.

^a Parenteral refers to administration by injection.

^b Pethidine in many countries.

^c Duration of antagonist activity.

^d For treatment of alcoholism.

Modified from Brody's Human Pharmacology Seventh Edition, 2025

<https://www.clinicalkey.com/#!/content/book/3-s2.0-B9780323846738000286>

Opioid pharmacokinetics

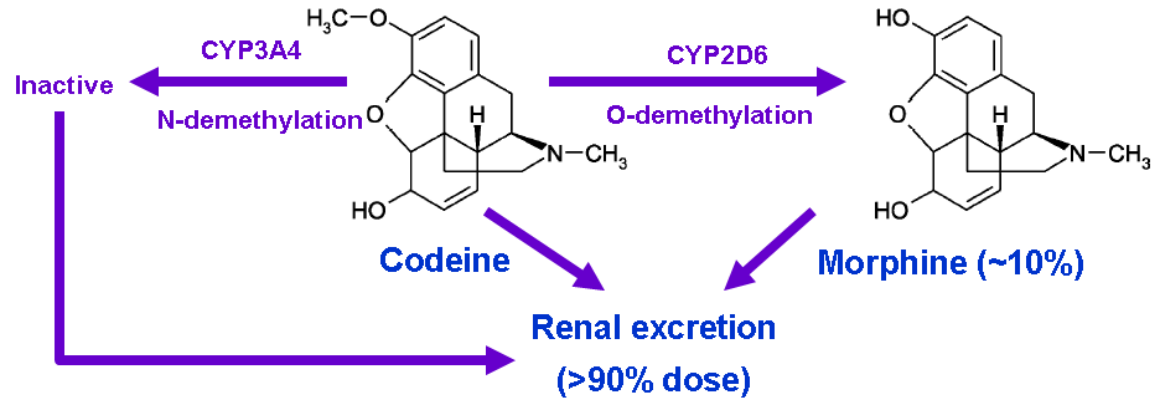
★ Half-life

- ◇ Immediate released forms of commonly used opioids have short half-lives and duration of action
- ◇ Methadone has very long $T_{1/2}$ 8-60 hr

★ Active forms of most commonly used opioids are excreted by kidney

Codeine

- ★ Metabolised by CYP2D6 to morphine
- ★ Dosing adjustment in patient with diminished renal function is similar to morphine
- ★ Max. dose 240-360 mg/day, initial 15-30 mg q 4-6 hr



CYP2D6 enzyme inhibitors
attenuate analgesic effects
of codeine

CYP2D6 polymorphisms
affect efficacy/safety

Tramadol

★ Mechanisms

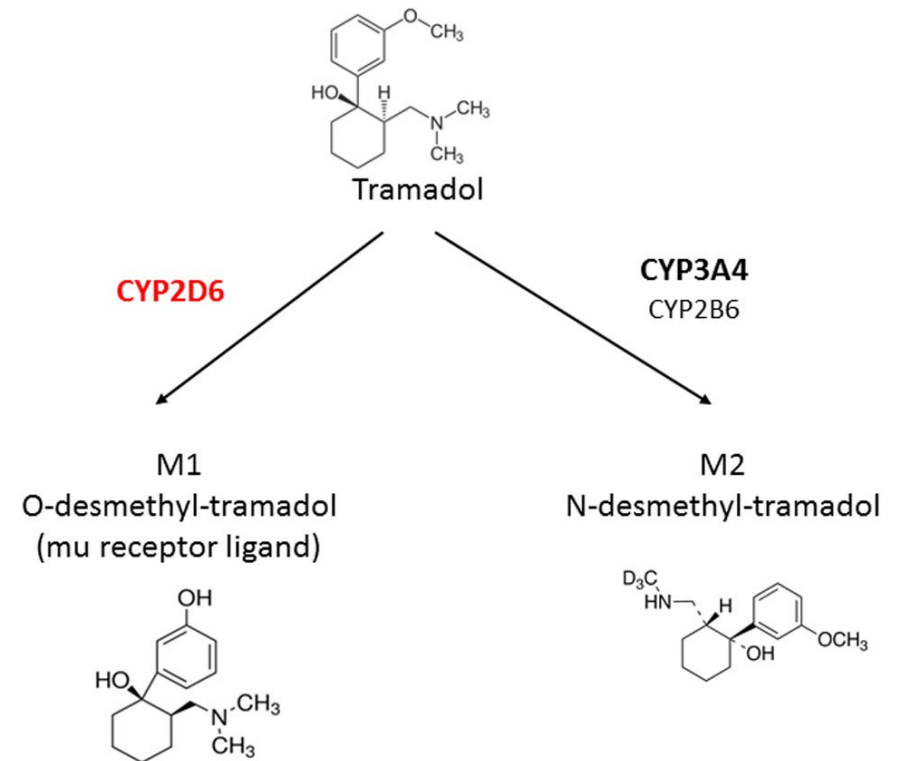
- ◇ μ receptor agonist
- ◇ NE transporter inhibitor
- ◇ 5-HT transporter inhibitor

★ Tramadol is metabolised by CYP2D6 to M1 metabolite which the affinity at μ receptor is close to morphine

★ Common side effects

- ◇ Nausea, vomiting, dizziness
- ◇ Palpitation
- ◇ Constipation

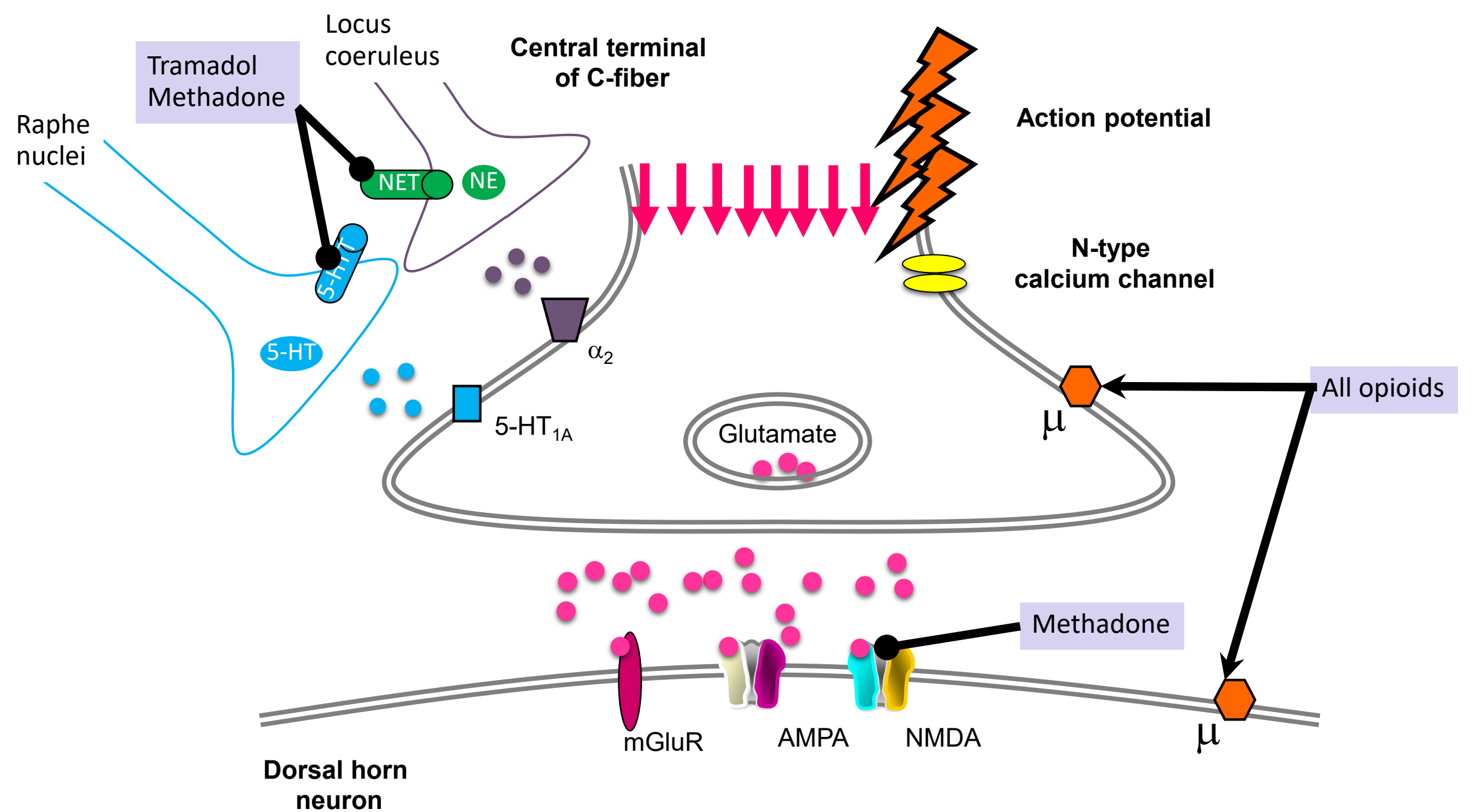
★ Reduce seizure threshold



Tramadol

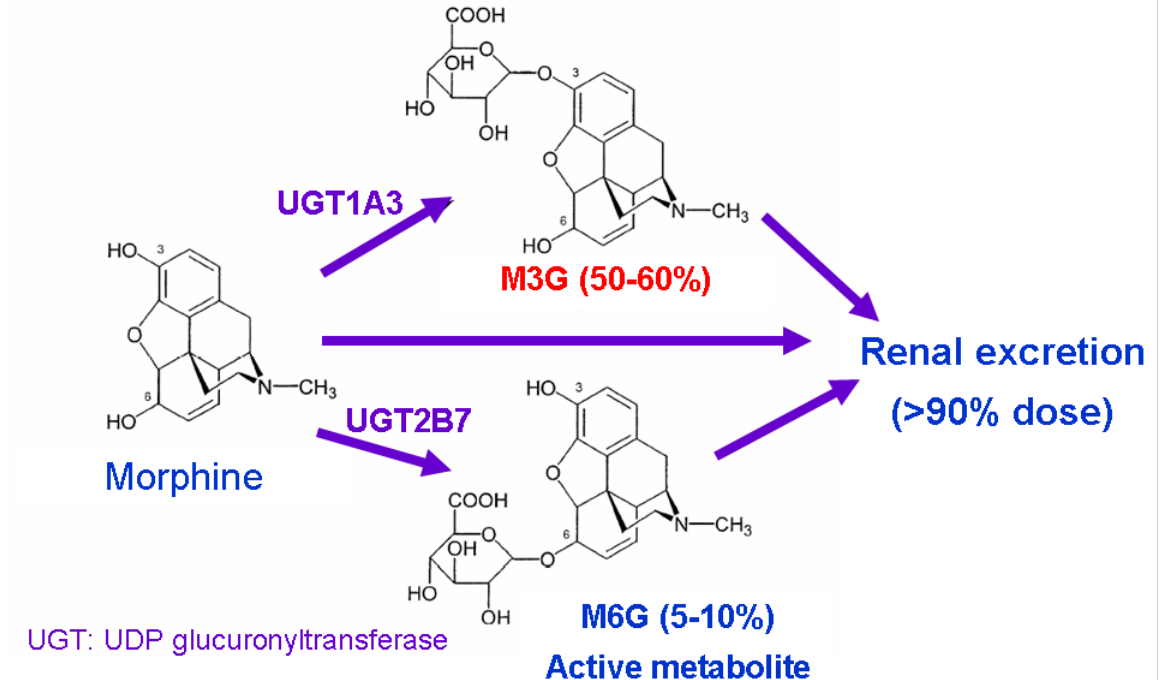
- ✱ Tramadol is excreted via kidney 30% as parent form and 60% as metabolites
- ✱ Mild to moderate renal impairment increases M1 level 20-40%

- Max dose 400 mg/day
- Dosing adjustment in renal impairment
 - Clcr 10-30 ml/min increase interval to 12 hours. The maximum recommended dose is 200 mg/day.
 - Clcr < 10 ml/min → 50 mg every 12 hours is recommended.
- Maximum recommended dose in geriatric patient is not exceed than 300 mg/day



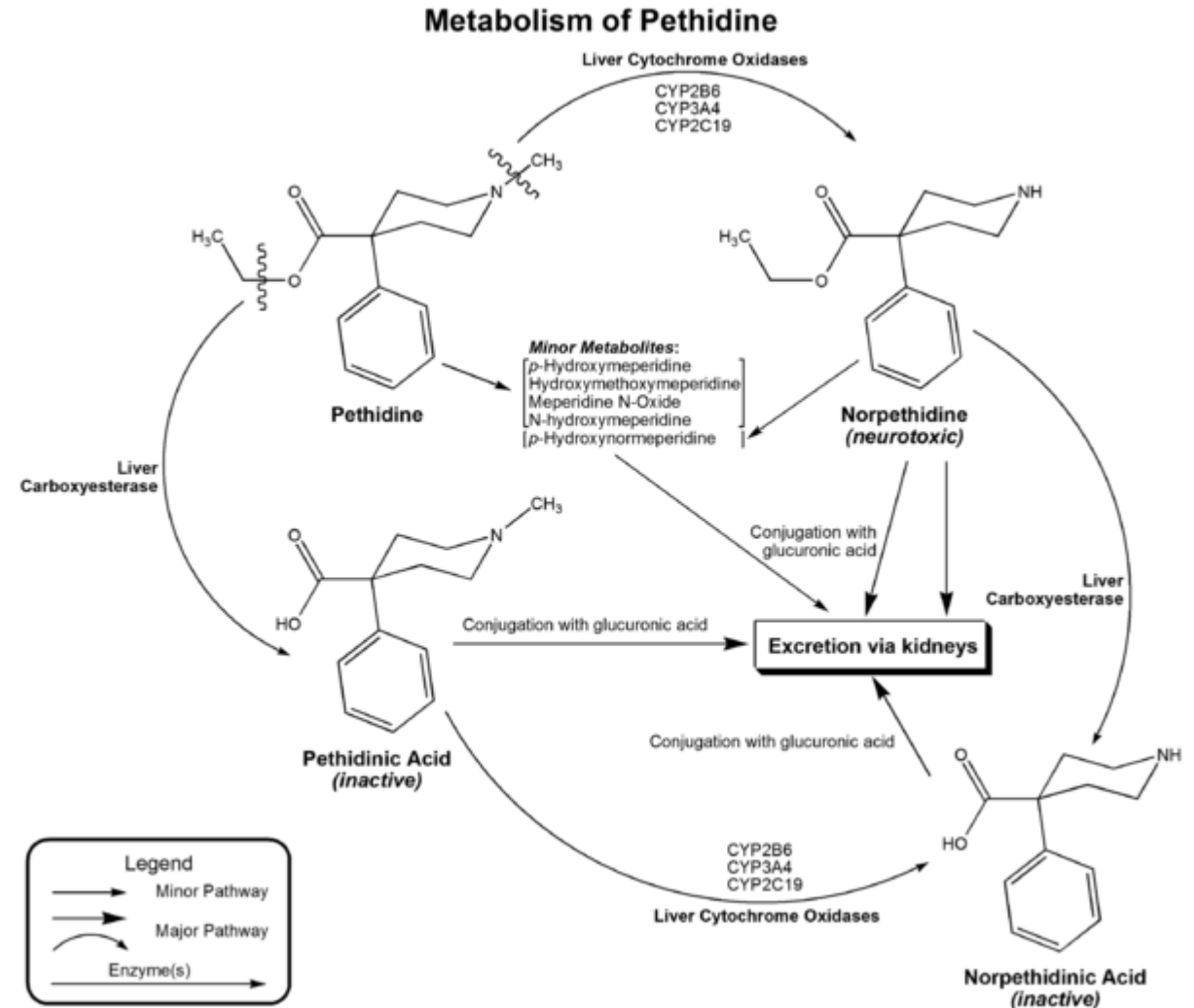
Morphine

- ★ Metabolised by UDP-glucuronosyl transferase (UGT) to
 - ◇ Morphine-3-glucuronide (M3G): neurotoxic and μ antagonist
 - ◇ Morphine-6-glucuronide (M6G): 40-50 more potent at μ
- ★ Almost of morphine and its metabolites are excreted by kidney



Pethidine (meperidine)

- ★ Pharmacologic effects
 - ◇ Rapid euphoric effect due to high lipophilicity
 - ◇ Anticholinergic
 - ◇ Inhibit 5-HTT
- ★ Metabolized to norpethidine (neurotoxic, $T_{1/2}$ 15-30 hours)
- ★ Excreted by kidney
- ★ Not for chronic use



Fentanyl

- ★ Very potent μ agonist
- ★ IV fentanyl has very fast onset (1 min) and short duration (30-60 min)
- ★ Fentanyl transdermal patch duration 3 days and $T_{1/2}$ after patch removal 13-22 hr
- ★ Metabolized by CYP3A4 to inactive metabolite
- ★ CYP3A4 inhibitors increase fentanyl level → fatal respiratory depression



Opioid metabolism, metabolites and excretion

Drug	Metabolism	Active Metabolites	Urine excretion
Weak opioids			
Codeine	CYP2D6	Yes	90%
Tramadol	CYP2D6	Yes	90%
Strong opioids			
Morphine	UGT	Yes	90%
Meperidine	CYP2B6, CYP3A4	Yes	70%
Oxycodone	CYP2D6 and CYP3A4	Yes	19-64%
Methadone	CYP2D6, CYP3A4, CYP1A2	No	<10%
Fentanyl	CYP3A4	No	75%
Partial Agonists and Mixed-Acting Compounds			
Buprenorphine	CYP3A4, UGT	No	30%
Nalbuphine	UGT and CYPs	No	

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4.7 Analgesics

4.7.1 Opioid analgesics

1.	Buprenorphine hydrochloride วัตถุที่ออกฤทธิ์ต่อจิตและประสาทประเภท 2	sublingual tab, sterile sol	ค
2.	Codeine phosphate ยาเสพติดให้โทษประเภท 2	tab	ค
3.	Fentanyl ยาเสพติดให้โทษประเภท 2 เจ็อนไซ ใช้กับผู้ป่วยโรคมะเร็งที่มีความเจ็บปวดรุนแรง	sterile sol (as citrate), transdermal therapeutic system (as base)	ค
4.	Methadone hydrochloride ยาเสพติดให้โทษประเภท 2	tab, oral sol	ค
5.	Morphine sulfate ยาเสพติดให้โทษประเภท 2	cap, tab, SR cap, SR tab, oral sol, sterile sol	ค
6.	Nalbuphine hydrochloride	sterile sol	ค
7.	Pethidine hydrochloride ยาเสพติดให้โทษประเภท 2	sterile sol	ค
8.	Tramadol hydrochloride	cap, tab, SR cap, SR tab, sterile sol	ค